Software Requirements Specification

for

NITC CCC Terminal Booking Tracker

Terminal Availability and Booking Request Management System

Version 1.1 – Revised Project Baseline

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Revision History

Author	Date	Description	Version
Project Team	2 Nov 2026	Initial SRS defining email-based reservation tracking, terminal availability and faculty booking requests.	1.0
Project Team	6 Nov 2026	Revised role permissions, clarified faculty request approval, booking cancellation and conflict handling.	1.1

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the requirements of the NITC CCC Terminal Booking Tracker. It is a web-based application for viewing the availability of terminal rooms managed by the Computer Centre (CCC) of the National Institute of Technology Calicut (NITC).

The system collects reservation information from the designated CCC mailbox and displays confirmed bookings on a common dashboard. Students can view the dashboard only. Faculty members can view the dashboard and submit requests for available terminal-room time slots. CCC administrators review faculty requests, approve or reject them, and can cancel confirmed bookings.

This document defines the expected behaviour, interfaces, constraints and acceptance conditions of the first release.

1.2 Document Conventions

The following conventions are used in this document:

- **REQ:** Identifier used for functional requirements.
- **Confirmed booking:** A reservation that is part of the official schedule and blocks the corresponding room and time interval.
- **Faculty request:** A request awaiting a CCC administrator's decision; it is not a confirmed booking.

1.3 Intended Audience

This document is intended for the project guide, project developers, testers, CCC management, faculty users and project evaluators. Developers can use it to implement the application, while testers can use it to verify the stated requirements.

1.4 Product Scope

The first release supports the six terminal rooms managed by CCC. It provides a common dashboard showing confirmed reservations and available predefined time slots.

The system has two sources of confirmed booking information:

- **CCC email reservations:** Reservation-related emails received in the designated CCC mailbox are retrieved. The system extracts and validates booking details before adding a non-conflicting reservation to the schedule. The system does not perform sender verification because the designated mailbox is the source provided for CCC reservation emails.
- **Faculty booking requests:** Faculty members submit requests through the application for available predefined time slots. A request becomes a confirmed booking only after a CCC administrator approves it.

The main objectives are to:

- Display confirmed bookings and availability for the six configured terminal rooms.
- Reduce manual searching through CCC reservation emails.
- Allow faculty members to request available terminal-room time slots.
- Prevent overlapping confirmed bookings for the same room.
- Allow CCC administrators to approve or reject faculty requests and cancel confirmed bookings.
- Provide students with read-only access to the availability dashboard.

1.5 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
2. Sample CCC reservation email, subject to availability.
3. Applicable NITC and CCC policies, subject to confirmation by CCC management.
4. SRS First-Draft Review, NITC CCC Terminal Booking Tracker, reviewed 3 November 2026.

2 Overall Description

2.1 Product Perspective

The NITC CCC Terminal Booking Tracker works alongside the existing CCC reservation process. It provides a consolidated view of reservations received through the designated CCC mailbox and faculty requests approved by CCC administrators. It does not independently allocate rooms or replace CCC's authority over the official schedule.

The application retrieves reservation-related emails from the designated CCC mailbox, extracts and validates their booking details, checks for duplicate messages and schedule conflicts, and updates the booking records when appropriate. Faculty members use the application to request available time slots. CCC administrators make the final decision on faculty requests and can cancel confirmed bookings.

2.2 Product Functions

The system shall provide the following main functions:

- Retrieve reservation-related emails from the designated CCC mailbox and extract required booking details without performing sender verification.
- Validate extracted email details, detect duplicate messages and identify booking conflicts.
- Display the six terminal rooms, confirmed bookings and available predefined time slots on a common dashboard.
- Allow students to view the dashboard without booking or management permissions.
- Allow faculty members to view the dashboard, submit requests for available slots and view the status of their own requests.
- Allow CCC administrators to review, approve or reject faculty requests and cancel confirmed bookings.
- Display email records that require administrator attention without automatically changing an existing confirmed booking.

2.3 User Classes and Characteristics

User Class	Description	Permissions
Student	Uses the application to check terminal-room availability and confirmed bookings.	View dashboard and room schedule only. Cannot submit requests, view faculty request records, approve, reject or cancel bookings.
Faculty	Needs a terminal room for an academic activity, assessment or other permitted purpose.	View dashboard, submit requests for available predefined slots and view the status of own requests. Cannot approve or cancel confirmed bookings.
CCC Administrator	Manages faculty requests and the confirmed terminal-room schedule.	View dashboard, view pending requests, approve or reject requests, view email records requiring review and cancel confirmed bookings.

2.4 Operating Environment

- The application shall be accessible through a web browser on desktop, tablet and mobile devices.
- The application shall run on a server or virtual machine approved for the project.
- The application shall connect to the designated CCC mailbox using an access method approved for the project.
- A persistent database shall store room, email, booking, request, user and audit information.
- All booking dates and times shall use Indian Standard Time (IST).

2.5 Design and Implementation Constraints

- The first release shall support the six terminal rooms configured for deployment.
- Faculty requests shall use the predefined time slots configured by CCC.
- Only CCC administrators shall approve or reject faculty booking requests.
- Students shall have dashboard-only, read-only access.
- A faculty request shall not become a confirmed booking until it is approved by a CCC administrator.
- The designated CCC mailbox shall be the source of email reservation records used by the application.

2.6 Assumptions and Dependencies

Assumptions

- CCC reservation-related emails are delivered to a designated mailbox accessible to the application.
- The designated mailbox is the source of CCC reservation emails; the application is not required to verify the sender address.
- Reservation emails contain, or can be interpreted to provide, the room identifier, reservation date, start time, end time and purpose.
- Faculty members and CCC administrators have access to an NITC institutional identity or an institution-approved authentication method.
- CCC provides the six room identifiers, predefined time slots and applicable booking rules.

Dependencies

- Access to the designated CCC mailbox or representative reservation email samples.
- Confirmation of the six room names and identifiers.
- Confirmation of predefined time slots, room operating hours and any booking restrictions.
- Availability of an approved authentication mechanism for faculty members and CCC administrators.
- Confirmation of booking advance-period and data-retention policies.

3 External Interface Requirements

3.1 User Interfaces

The application shall provide the following interfaces:

- **Availability Dashboard:** Displays the six terminal rooms, confirmed bookings and available predefined time slots. Students, faculty members and CCC administrators can access this interface.
- **Faculty Request Form:** Allows an authenticated faculty member to select an available room, date and predefined time slot(s), enter a purpose and submit a request.
- **My Requests:** Allows a faculty member to view only the requests submitted by that faculty member and their current statuses.
- **CCC Administration:** Allows CCC administrators to view pending faculty requests, approve or reject them, view email records requiring review and cancel confirmed bookings.
- **Booking Details:** Displays the room, date, start time, end time, purpose and booking status for a confirmed booking.

The interface shall use clear labels for booking and request statuses and shall remain usable on common desktop and mobile screen sizes.

3.2 Hardware Interfaces

The application has no direct hardware interface. Users access it through computers, tablets or smartphones with a supported web browser. The application requires server infrastructure approved for deployment.

3.3 Software Interfaces

Interface	System or Data	Purpose
CCC Email Service	Designated CCC mailbox	Supplies reservation-related emails for terminal-room booking records.
Application Database	Persistent database	Stores room, email, booking, request, user and audit records.
Web Browser	Supported browser	Displays the dashboard and role-specific pages.
Authentication Service	NITC identity service or approved login mechanism	Authenticates faculty members and CCC administrators.

3.4 Communications Interfaces

- Browser-to-server communication shall use HTTPS in the approved deployment.
- Mailbox communication shall use an authenticated and encrypted connection supported by the mail provider.
- The application shall check the designated CCC mailbox at intervals of two minutes.
- The dashboard shall display the last successful mailbox synchronization time and refresh after confirmed booking updates.

4 System Features

4.1 CCC Email Booking Management

4.1.1 Description and Priority

This feature retrieves reservation-related emails from the designated CCC mailbox, extracts booking information, validates the extracted details and updates the booking schedule when the email represents a valid, non-conflicting reservation. Sender verification is not part of this system because the designated CCC mailbox is the configured source of reservation emails. This is a high-priority feature.

4.1.2 Functional Requirements

1. **REQ-1:** The system shall retrieve new reservation-related emails from the designated CCC mailbox every two minutes without performing sender-address verification.
2. **REQ-2:** The system shall extract and validate the room identifier, reservation date, start time, end time and purpose from each supported reservation email.
3. **REQ-3:** The system shall identify previously processed messages and prevent duplicate booking creation using the available unique message identifier.
4. **REQ-4:** The system shall store valid, non-conflicting email reservations as Confirmed and mark incomplete, invalid, ambiguous or conflicting emails as Needs Review without automatically changing existing confirmed bookings.

4.2 Terminal Availability Dashboard

4.2.1 Description and Priority

This feature displays confirmed booking information and available predefined time slots for the six configured terminal rooms. Students, faculty members and CCC administrators can view the dashboard. Students have no booking or management permissions. This is a high-priority feature.

4.2.2 Functional Requirements

1. **REQ-5:** The system shall display all six configured terminal rooms and their confirmed bookings for a selected date.
2. **REQ-6:** The system shall display the room, date, start time, end time and purpose of each confirmed booking.

3. **REQ-7:** The system shall mark a predefined time slot as Booked if any part of it overlaps a confirmed booking for the same room. Pending requests shall not block availability.
4. **REQ-8:** The system shall display the last successful mailbox synchronization time and show a warning if synchronization has not succeeded within the last 15 minutes.

4.3 Faculty Booking Request

4.3.1 Description and Priority

This feature allows authenticated faculty members to submit booking requests for available terminal-room time slots. The request remains Pending until a CCC administrator approves or rejects it. A request becomes a confirmed booking only after approval and a final conflict check. This is a high-priority feature.

4.3.2 Functional Requirements

1. **REQ-9:** The system shall allow authenticated faculty members to select a room, date, one or more available predefined time slots and a booking purpose.
2. **REQ-10:** The system shall validate the selected slots against confirmed bookings and prevent requests for conflicting slots or past dates.
3. **REQ-11:** The system shall save a successfully submitted faculty request with Pending status. A Pending request shall not be treated as a confirmed booking or block availability.
4. **REQ-12:** The system shall allow faculty members to view the details and current status of only their own requests, without permission to approve, reject or cancel bookings.

4.4 CCC Administrator Management

4.4.1 Description and Priority

This feature allows CCC administrators to make decisions on faculty booking requests and maintain the confirmed booking schedule. CCC administrators retain final authority over whether a faculty request is accepted. An administrator can approve or reject a pending request and cancel a confirmed booking. This is a high-priority feature.

4.4.2 Functional Requirements

1. **REQ-13:** The system shall allow authenticated CCC administrators to view pending faculty booking requests and approve or reject them.
2. **REQ-14:** Before approval, the system shall perform a final conflict check and create a Confirmed booking only if no overlapping confirmed booking exists.
3. **REQ-15:** The system shall allow CCC administrators to cancel existing confirmed bookings and update the dashboard availability accordingly.
4. **REQ-16:** The system shall record administrator approvals, rejections and cancellations in the audit records and prevent simultaneous approvals from creating overlapping confirmed bookings.

5 Other Nonfunctional Requirements

5.1 Performance

- The dashboard shall load within five seconds for up to 20 simultaneous users under the approved project test environment.
- The system shall check the CCC mailbox every two minutes and process or flag retrieved messages within two minutes of retrieval, excluding mailbox-provider outages.
- The dashboard shall reflect an approved, rejected or cancelled booking action within 60 seconds of a successful database update.

5.2 Reliability

- A temporary mailbox failure shall not delete previously stored booking records. The system shall retry synchronization during a later scheduled check.

- Failed email retrieval or parsing attempts shall be logged with their time and processing result.
- The system shall preserve stored records after an application restart.
- The system shall prevent simultaneous approval actions from creating overlapping confirmed bookings.
- If mailbox synchronization fails, the dashboard shall continue to display previously stored confirmed bookings and show the last successful synchronization time and stale-data warning.

5.3 Security

- Faculty and CCC administrator functions shall require authentication.
- The server shall verify the user's role before allowing a faculty request or administrator action.
- Students shall have read-only dashboard access and shall not be able to access faculty request or administrator functions.
- CCC mailbox credentials and application secrets shall not be stored in public source code.
- The dashboard shall not expose mailbox credentials or complete private email content.
- Browser-to-server and mailbox communication shall use encrypted connections in the approved deployment.

5.4 Software Quality Attributes

- **Maintainability:** Booking, mailbox processing, faculty requests and dashboard functions shall have separate, identifiable responsibilities.
- **Scalability:** The system shall allow additional users and booking records without changing the basic booking workflow.
- **Portability:** The application shall run in the approved server environment and support the browsers selected for project testing.
- **Testability:** The system shall maintain identifiable processing results and logs so that email, booking and administrator operations can be verified.

5.5 Business Rules

- Students can view the dashboard only. They cannot submit booking requests or perform booking-management actions.
- Faculty members can view the dashboard and submit requests only for available predefined time slots.
- Only CCC administrators can approve or reject faculty booking requests and cancel confirmed bookings.
- A faculty request remains Pending until an administrator makes a decision. A Pending request does not reserve or block a time slot.
- A faculty request becomes a Confirmed booking only after administrator approval and a successful final conflict check.
- A terminal room shall not have two overlapping Confirmed bookings.
- Reservation-related messages received in the designated CCC mailbox are processed without sender-address verification.
- An incomplete, ambiguous or conflicting email shall be marked Needs Review and shall not automatically overwrite or cancel an existing Confirmed booking.
- All booking dates and times shall be maintained and displayed in IST. A confirmed booking that overlaps any part of a predefined slot makes that slot Booked.
- Faculty members shall not submit requests for past dates.

6 Other Requirements

6.1 Data Requirements

The system shall maintain the following main data records:

- **Terminal Room:** Room identifier and room name.
- **Email Record:** Source message identifier, subject or reference, received timestamp, extracted booking fields and processing status. Sender verification information is not required.
- **Booking:** Booking identifier, room, date, start time, end time, purpose, source and booking status.
- **Booking Request:** Request identifier, faculty identifier, room, date, selected slot(s), purpose, submission timestamp and request status.
- **User:** User identifier, institutional identifier and role for authenticated faculty and CCC administrator accounts.
- **Audit Record:** Administrator identity, action, affected record identifier and timestamp.
- **Synchronization Record:** Last successful mailbox check time and the result of the most recent synchronization attempt.

The system shall retain a reference to the source message for each email-derived booking. The system shall not store unrelated mailbox content. A faculty booking request and the Confirmed booking created after approval shall be linked so that the request's decision can be traced.

6.2 Status Definitions

The system shall maintain separate statuses for faculty booking requests, confirmed bookings and email records.

Booking Request Status

- **Pending:** Submitted by faculty and awaiting an administrator decision. It does not block the slot.
- **Approved:** Accepted by a CCC administrator; a Confirmed booking is created.
- **Rejected:** Declined by a CCC administrator; no Confirmed booking is created.

Booking Status

- **Confirmed:** A reservation accepted through a valid CCC mailbox record or an approved faculty request.
- **Cancelled:** A previously Confirmed booking cancelled by a CCC administrator. A cancelled booking no longer blocks availability.

Email Record Status

- **Processed:** The email was retrieved, validated and handled successfully.
- **Needs Review:** The email is incomplete, invalid, ambiguous or conflicts with an existing confirmed booking.
- **Duplicate:** The same source message has already been processed and shall not create another booking.
- **Failed:** The system could not retrieve or process the email because of a technical or parsing failure.

6.3 Booking Conflict Rules

- A conflict exists when two time intervals for the same room overlap for a non-zero duration.
- Two bookings are considered overlapping when $Start_A < End_B$ and $Start_B < End_A$.
- A booking ending exactly when another booking starts does not create an overlap.
- Only Confirmed bookings block faculty requests and appear as Booked on the dashboard. Pending requests do not block slots.
- A new email reservation that conflicts with an existing Confirmed booking shall be marked Needs Review. The system shall preserve the existing Confirmed booking until an administrator takes an appropriate action.
- The system shall perform a final conflict check immediately before confirming a faculty request.

6.4 Items Requiring CCC Confirmation

The following operational details shall be confirmed before final deployment:

- The six terminal room identifiers and display names.

- The predefined time slots and room operating hours.
- The supported format or formats of CCC reservation emails.
- How a reservation email identifies a change or cancellation to an earlier booking, if such messages are included in the first release.
- The policy to follow when two CCC reservation emails contain conflicting booking details.
- The maximum advance period for faculty requests and the required data-retention period.

A Glossary

Term	**Definition**
CCC	Computer Centre of the National Institute of Technology Calicut.
NITC	National Institute of Technology Calicut.
Terminal Room	A computer laboratory or room managed by CCC and available for scheduled use.
Dashboard	The common page that displays terminal-room bookings and availability.
Booking	A confirmed reservation of a terminal room for a date and time interval.
Booking Request	A request submitted by a faculty member that requires CCC administrator approval.
Pending	A faculty booking request awaiting an administrator decision; it does not block availability.
Approved	A faculty booking request accepted by CCC; a Confirmed booking is created.
Rejected	A faculty booking request declined by CCC; no Confirmed booking is created.
Confirmed	A reservation accepted through a valid CCC mailbox record or an approved faculty request.
Cancelled	A previously confirmed booking cancelled by a CCC administrator.
Needs Review	An email record requiring administrator attention before it can change the confirmed schedule.
Overlap	Two time intervals for the same room that share a non-zero period.
Predefined Slot	A time interval configured by CCC and offered in the faculty request form.
IST	Indian Standard Time (UTC+05:30).
Audit Record	A record of an administrator action affecting a request or booking.

B Analysis Models

B.1 Use Case Summary

Use Case	Actor
View terminal availability	Student, Faculty, CCC Admin
View confirmed bookings for a selected date and room	Student, Faculty, CCC Admin
Submit a request for an available terminal slot	Faculty
UView own booking request status	Faculty
Review pending faculty requests	CCC Admin
Approve or reject a faculty request	CCC Admin
Cancel a confirmed booking	CCC Admin
Retrieve and process reservation emails from the designated CCC mailbox	System
View email records requiring review	CCC Admin

B.2 Faculty Booking Request Flow

1. The faculty member opens the dashboard.
2. The faculty member selects a room and date. The system displays available predefined time slots based on confirmed bookings.
3. The faculty member selects one or more available slots and enters the booking purpose.
4. The system checks the selected slots again. If a confirmed booking overlaps the selection, submission is blocked.
5. If no conflict exists, the system saves the request with Pending status. The pending request does not block the slot.
6. A CCC administrator reviews the request and performs a final conflict check.
7. If approved, the system creates a Confirmed booking and updates the dashboard. If rejected, no booking is created.
8. The faculty member can view the decision in My Requests.

B.3 CCC Email Processing Flow

1. The system checks the designated CCC mailbox every two minutes and retrieves new reservation-related messages.
2. The system checks whether each message has already been processed. No sender-address verification is performed.
3. The system extracts the room, date, start time, end time and purpose.
4. The system validates the extracted fields and checks for a conflict with existing confirmed bookings.
5. If the message is valid, not a duplicate and has no conflict, the system creates a Confirmed booking and updates the dashboard.
6. If the message is incomplete, invalid, ambiguous or conflicting, the system marks it Needs Review. Duplicate messages are identified and do not create another booking.

B.4 Role and Permission Summary

Function	Student	Faculty	CCC Admin
View dashboard	Yes	Yes	Yes
View confirmed room schedule	Yes	Yes	Yes
Submit booking request	No	Yes	No
View own requests	No	Yes	No
Approve or reject faculty requests	No	No	Yes
Cancel confirmed booking	No	No	Yes
View email records requiring review	No	No	Yes